

Sean Dahlman

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SUMMARY

GIS professional with 6+ years of experience spanning municipal operations, environmental monitoring, and geospatial automation. Currently completing an MSc in GIS at Vrije Universiteit Amsterdam, with thesis research applying multi-modal deep learning to urban tree canopy detection. Growing specialization in GeoAI and remote sensing. US citizen with full work authorization, available to relocate.

EXPERIENCE

GIS Analyst | City of Boise, ID

December 2023 – October 2024

- Automated daily data validation workflows in Python and Bash, reducing processing time from 4 hours to 45 minutes (over 80% reduction).
- Designed and maintained internal and third-party geospatial data products supporting city operations for 250,000 residents.
- Performed complex geospatial analysis to generate statistics, discover correlations, and support city service and planning decisions.
- Maintained spatial assets within VueWorks CMMS; verified accuracy of field-collected spatial data and enforced city GIS standards.
- Managed relational databases, attribute data entry, and metadata for enterprise geospatial features.
- Provided GIS software support to city users, resolved end-user data problems, and authored training manuals and procedures.
- Participated in enterprise and department-level GIS planning and coordination meetings.

Biological Technician | USDA Forest Service

April 2018 – November 2023

- Conducted GIS analysis, data management, and data entry through Natural Resource Manager (NRM) via ArcGIS Pro.
- Performed Aerial Detection Surveying (ADS) from aircraft to map forest health conditions across multi-state regions.
- Executed traverse surveys using RTK Trimble GPS, theodolites, and electronic distance measuring systems.
- Managed spatial data for Threatened, Endangered, and Sensitive species surveys through NRM.
- Conducted vegetation habitat data analysis using Rocky Mountain Research Station standards.
- Led crews of seasonal employees in field data collection; wrote and secured two environmental grants.
- Assisted with drone flights for vegetation surveying; maintained technical reports, grants, and agreements.

Web Developer (Part-time / Weekend Contractor) | Somerset Pacific

January 2019 – Present

- Designed and maintain 20+ corporate websites on a weekend/part-time basis (approximately 5 hours per week).
- Perform routine updates, troubleshoot issues, and optimize site performance for SEO and mobile compatibility.
- Manage DNS, AWS, Webflow, and SSL certificates; coordinate with stakeholders on site changes and branding.

SELECTED PROJECTS

GeoAI Urban Tree Canopy Detection — MSc thesis research, 2025

- Developed multi-modal U-Net deep learning pipeline fusing NAIP 60cm imagery (RGB+NIR) with LiDAR-derived nDSM for city-wide tree canopy detection across 227 km² of Boise, Idaho.
- Achieved test IoU of 0.952 and F1 of 0.975; adding LiDAR height as a model input channel improved IoU by 30% over spectral-only methods.
- Created physics-based training label generation workflow (NDVI + nDSM thresholds), validated against 34,499 field-surveyed trees at 90% detection rate, eliminating need for manual annotation.
- Delineated 261,262 individual tree crowns via marker-controlled watershed segmentation; exported as GeoTIFFs and shapefiles for urban forestry applications.
- Tools: PyTorch, segmentation_models_pytorch, PDAL/laspy, GDAL/Rasterio, Three.js.

Prescribed Burn Effectiveness: Digital Twin Framework — MSc project, 2025

- Simulated prescribed burn effects on the 2021 Snake River Complex Fire using FlamMap fire behavior modeling; treated scenario eliminated crown fire activity and reduced flame length 55%, fireline intensity 70%.
- Validated model outputs against observed burn severity via Sentinel-2 dNBR analysis in Google Earth Engine.
- Built 3D terrain visualizations in Blender from USGS 3DEP DEM with GEE-derived tree locations.
- Tools: FlamMap, Google Earth Engine, Python (NumPy, rasterio, SciPy), Blender, ArcGIS Pro.

Predictive Invasive Weed Mapping in Timber Harvest Zones — *MSc project, 2025*

- Synthesized USFS invasive weed surveys, timber harvest zones, and DEM data across 57 Ranger Districts to predict invasive species establishment.
- Random Forest classifier achieved AUC 0.987 with minimal overfitting; DBSCAN outlier detection applied for data quality control.
- Identified that 88.5% of sampled locations require enhanced weed management post-harvest.
- Tools: Python (scikit-learn, DBSCAN, K-Means, GMM), ArcGIS Pro, USFS FSGeodata.

Quantifying Farmland Decline in Sardinia — *MSc project, 2025*

- Multi-temporal Landsat 5/8/9 NDVI and SAVI analysis across all Sardinian provinces (2007–2024).
- Post-classification comparison identified transitions between agriculture, forest/orchard, urban, and bare/fallow land cover classes.
- Documented 67.9% NDVI increase on registered farm plots, indicating agricultural abandonment and reforestation rather than active farming.
- Tools: Google Earth Engine, ArcGIS Pro, Landsat 5/8/9, ISTAT data.

Flood Impact Assessment — Rockhampton, Queensland — *MSc project, 2025*

- Python-based raster analysis pipeline using rioarray, geopandas, and xarray to identify flood extent from Sentinel-2 NDWI imagery.
- Multi-temporal comparison (peak flood, recovery, baseline) quantified 67,270 hectares of flooding.
- Integrated OSM infrastructure and JRC GHS building/population rasters to assess exposure of roads, power lines, and built-up areas.
- Tools: Python (rioarray, geopandas, xarray, matplotlib), Sentinel-2, OpenStreetMap, JRC GHS.

EDUCATION

MSc, Geographic Information Systems | Vrije Universiteit Amsterdam 2024 – Present (thesis stage)

- Coursework complete. Current thesis: multi-modal deep learning for urban tree canopy detection.
- Relevant coursework: GIS, Remote Sensing, GeoAI, Database Management, Geospatial Analysis, Environmental Science.

BA, Conservation Biology | The College of Idaho 2014

- Relevant coursework: Geographic Information Systems, Statistics, Botanical Surveying.

SKILLS

GIS & Remote Sensing: ArcGIS Pro, ArcGIS Enterprise, ArcGIS Online, ArcMap, QGIS, Google Earth Engine, Natural Resource Manager (NRM), FlamMap

GeoAI & Machine Learning: PyTorch, segmentation_models_pytorch, U-Net, DeepLabV3+, scikit-learn, object detection, semantic segmentation, Random Forest

Programming & Automation: Python (ArcPy, GeoPandas, rioarray, xarray, rasterio, NumPy, Pandas), SQL, R, JavaScript, Bash

Databases: PostgreSQL/PostGIS, relational database design, spatial data modeling, metadata documentation (FGDC/ISO)

Remote Sensing Data: Sentinel-2, Landsat 5/8/9, NAIP, LiDAR/nDSM, NDVI, NDWI, dNBR, OpenStreetMap

Field Survey & Data Collection: RTK Trimble GPS, theodolite and EDM surveying, Aerial Detection Surveying (ADS), drone-assisted vegetation surveys

Infrastructure: AWS, VPS management, DNS and SSL administration, Docker

CERTIFICATIONS

Aerial Crewman Training — USDA Forest Service